

DAVID XU

Phone: (+1) 510 365 6892 / (+064) 022 425 6280 Email: rav4@berkeley.edu [GitHub](#)

EDUCATION

University of California, Berkeley

(Expected May 2027)

Bachelor of Science - Electrical Engineering and Computer Science (EECS)

- Coursework: Operating Systems and System Programming • Optimization Models • Introduction to Machine Learning • Programming Languages/Compilers • Efficient Algorithms and Intractable Problems

PROJECTS & RESEARCH

Multimodal Contrastive Learning for Cancer Cell Identification

Spring 2026

- Co-authored a paper accepted to NeurIPS 2026 Workshop on **Multimodal Foundation Models for Life Sciences**.
- Developed a multimodal contrastive learning framework for alternative **Differential Expression Analysis**
- Integrated transcriptomic features with representation learning to identify early malignant transition

Risk Communication Analysis of Drug Labels using Natural Language Inference

Summer 2026

- Created framework for **longitudinal analysis** of risk communication for drug labels using **Natural Language Inference**
- Presented findings, discussions and pipeline in paper under submission for LREC 2026.
- Used **Zero-shot** text classification methods in **Natural Language Processing**.

Battlecode 2025 (MIT)

Spring 2025

- Placed 9th overall in MIT's longest-running autonomous **robotics & programming** competition.
- Designed **pathfinding** and **decision-making systems** under adversarial, partially observable environments
- Implemented **efficient traversal, environment detection, and strategic resource allocation**

Measuring Information Safety in LLMs

Fall 2025

- Feature design inspired by computational social science literature
- Designed a **benchmarking** framework to measure political bias and influence in **large language models**
- Built backend infrastructure for LLM-Arena style LLM-versus-LLM evaluation using Supabase

WORK EXPERIENCE:

North Shore Community Trust

Spring 2023, Summer 2024, 2025

Program Director Lead

Auckland, NZ

- Led recruitment, sponsorship outreach, and program design for a regional education initiative
- Managed operations for programs serving 50+ students **annually** with a **7-person instructional staff**
- Authored curriculum materials and coordinated logistics across multiple cohorts

Ahoura Center for Smart Energy Systems

Summer 2023

Research Intern

Auckland, NZ

- Conducted **literature review** on energy system design aligned with national carbon neutrality goals.
- Proposed process-level design improvements for smart energy infrastructure.
- Collaborated with researchers on feasibility analysis and technical documentation.

SKILLS

Languages: Python · C/C++ · Java · Go · R · Ruby · HTML/CSS · JavaScript ML/Data: PyTorch · Hugging Face Transformers · NumPy · pytorch · pandas · scikit-learn · BeautifulSoup

Frameworks & Tools: React · Git · Supabase · Linux/shell

Areas: Machine Learning · NLP · Deep Learning · Systems Programming · Algorithms · Full-Stack Development

LEADERSHIP & SERVICE

Northcote Community After School Program

Summer 2021-Present

Head Tutor

Auckland, NZ

- Tutored students (Grades 5–11) in Algebra and Python, 6 hrs/week
- Designed individualized lesson plans and mentored students in foundational programming concepts

SimpleKeyboard - Co-founder

- Designed and developed an accessible keyboard application for children with special needs
- Led UX design with emphasis on accessibility, error tolerance, and diverse user behaviors

Auckland Community Service

- Provided one-on-one technical assistance to elderly community members
- Organized community events including information sessions and social gatherings

Berkeley DNA sequencing facility

- Gained hands-on experience with high-throughput sequencing workflows and in a shared research environment
- Worked within a remote, multi-user Linux environment, managing file systems, job scheduling.

ADDITIONAL INFORMATION

I am Authorized to work in New Zealand (Permanent Resident). I require CPT sponsorship for U.S. employment. Open to on-site (Bay Area, NZ, Australia) or remote roles.